Module 9: Inheritance Genetics — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain Mendel's Laws of Segregation and Independent Assortment
2. Predict outcomes of monohybrid and dihybrid crosses using Punnett squares
3. Distinguish between complete dominance, incomplete dominance, and codominance
4. Solve genetics problems involving multiple alleles (e.g., ABO blood types)
5. Analyze pedigrees to determine inheritance patterns (autosomal vs. sex-linked, dominant vs. recessive)
6. Explain polygenic inheritance and how it differs from single-gene traits
7. Describe how environment interacts with genotype to produce phenotype

Key Terms to Know

Mendelian Genetics

- **Gene** — A heritable unit of information; a segment of DNA encoding a protein or functional RNA
- **Allele** — An alternative form of a gene at a specific locus (e.g., "B" for brown eyes, "b" for blue eyes)
- **Locus** — The specific position of a gene on a chromosome
- **Genotype** — The combination of alleles an organism carries (e.g., Bb)
- **Phenotype** — The observable trait expressed (e.g., brown eyes)
- **Homozygous** — Having two identical alleles (AA or aa)
- **Heterozygous** — Having two different alleles (Aa)
- **Dominant** — The allele expressed in the heterozygote (masks the recessive allele)
- **Recessive** — The allele only expressed when homozygous (masked by dominant in heterozygote)
- **True-Breeding** — An organism homozygous for a trait (always produces offspring with same phenotype)

- **P_0 Generation** — The parental generation in a cross
- **F_1 Generation** — First filial generation (offspring of P_0)
- **F_2 Generation** — Second filial generation (offspring of $F_1 \times F_1$)
- **Carrier** — A heterozygous individual who carries a recessive allele without expressing it

Mendel's Laws

- **Law of Segregation** — During gamete formation, the two alleles for a gene separate so that each gamete carries only one allele
- **Law of Independent Assortment** — Genes on different chromosomes assort independently during meiosis (applies to unlinked genes)
- **Test Cross** — Crossing an individual of unknown genotype (dominant phenotype) with a homozygous recessive (aa) to determine if it is AA or Aa

Beyond Simple Dominance

- **Incomplete Dominance** — Heterozygote shows a blended intermediate phenotype (e.g., red × white → pink flowers)
- **Codominance** — Both alleles are fully expressed in the heterozygote (e.g., AB blood type — both A and B antigens present)
- **Multiple Alleles** — More than two alleles exist for a gene in the population (e.g., ABO blood: I^A, I^B, i)
- **Pleiotropy** — One gene affects multiple phenotypic traits (e.g., sickle cell allele affects blood, spleen, oxygen transport)
- **Epistasis** — One gene modifies or masks the expression of another gene (e.g., coat color in Labrador retrievers)
- **Sex-Linked Inheritance** — Genes on the X chromosome; males (XY) are hemizygous — one copy determines phenotype
- **X-Linked Recessive** — Affected males: X^aY; carrier females: X^AX^a; affected females: X^aX^a (rare)

Polygenic Inheritance

- **Polygenic Trait** — A trait controlled by two or more genes, each contributing a small effect (e.g., height, skin color, weight)
- **Continuous Variation** — Polygenic traits show a range of phenotypes, often forming a bell curve distribution
- **Multifactorial** — Traits influenced by multiple genes AND environmental factors
- **Heritability** — The proportion of phenotypic variation in a population that is due to genetic variation

Classic Mendelian Ratios

Cross Type	Cross	Phenotypic Ratio
Monohybrid	Aa × Aa	**3:1** (3 dominant : 1 recessive)
Monohybrid (genotypic)	Aa × Aa	1:2:1 (1 AA : 2 Aa : 1 aa)
Test Cross	Aa × aa	1:1 (1 dominant : 1 recessive)
Dihybrid	AaBb × AaBb	**9:3:3:1**
Incomplete Dominance	Rr × Rr	1:2:1 (1 red : 2 pink : 1 white)

ABO Blood Type: A Model of Multiple Alleles + Codominance

Genotype	Phenotype (Blood Type)	Antigens on RBCs
I^AI^A or I^Ai	Type A	A antigen
I^BI^B or I^Bi	Type B	B antigen
I^AI^B	Type AB (codominance)	Both A and B
ii	Type O	Neither

Pedigree Analysis Tips

- **Autosomal Dominant**: Affected individuals appear in every generation; unaffected parents do NOT have affected children (usually)
- **Autosomal Recessive**: Can skip generations; two carrier parents → ¼ chance of affected child
- **X-Linked Recessive**: More males affected; affected fathers pass carrier status to ALL daughters; carrier mothers pass to ½ sons

Polygenic Inheritance: How Complex Traits Work

- Single-gene traits produce **discrete categories** (e.g., pea: tall or short)
- Polygenic traits produce **continuous variation** along a spectrum (e.g., human height forms a bell curve)
- More genes involved → smoother the distribution curve
- **Environment modifies the phenotype**: nutrition affects height, sunlight affects skin color, exercise affects weight
- **Genotype sets the range; environment determines where within that range an individual falls**

Example: Skin Color

- At least 3-4 genes contribute to human skin pigmentation
- Each gene has alleles that add to or reduce melanin production
- People with more "additive" alleles have darker skin
- Environment (UV exposure) further modifies pigmentation within the genetic range

Study Tips

1. **Practice Punnett squares** — Do them over and over until they're automatic
2. **Know the ratios** — 3:1 (monohybrid), 9:3:3:1 (dihybrid), 1:2:1 (incomplete dominance)
3. **Blood types are a favorite exam topic** — Master the I^A, I^B, i system

4. **Pedigree practice** — Work through pedigrees asking: dominant or recessive? Autosomal or sex-linked?
5. **Polygenic ≠ Mendelian** — Polygenic traits have continuous variation, not simple dominant/recessive ratios
6. **Genotype ≠ Phenotype** — Remember that environment modifies expression (e.g., hydrangea color changes with soil pH)
7. **Connect to Module 8** — Segregation happens during Meiosis I; independent assortment happens at Metaphase I